

Day 18 – Performance Benchmark & Profiling Report

Objective

Measure the performance of the Zecpath AI resume parsing pipeline and identify the stages that require optimization.

Test Resume

File: CAND-82624710.pdf

Pages: 2

Extracted Characters: 5305

Performance Benchmark

Resume Text Extraction

Metric	Result
Number of runs	5
Average time	0.4756 seconds
Minimum time	0.3323 seconds
Maximum time	0.6313 seconds

Complete Resume Parsing

Metric	Result
Number of runs	5
Average time	0.3501 seconds
Minimum time	0.3165 seconds
Maximum time	0.3976 seconds

Parser Stage Profiling

Pipeline Stage	Average Observed Time
PDF Text Extraction	~2.2 seconds during profiling run
Text Cleaning	~0.001 seconds
Section Detection	~0.001 seconds
Candidate Extraction	~0.001 seconds

The profiling results show that PDF processing is the dominant computational stage, while text cleaning, section detection, and candidate extraction require very little processing time.

Memory Usage

Peak memory: ~11.6 MB

The memory usage is acceptable for the current resume parsing workload.

Parser Output

Field	Result
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Candidate Name	SHIN AS PN
Skills	8
Experience	25
Education	2
Projects	27
Certifications	4

Candidate Name Verification

The parser returned: SHIN AS PN

This value was manually verified against the original PDF. The PDF itself contains the candidate name as: SHIN AS PN. Therefore, the parser output is correct and no correction was applied.

Findings

1. PDF extraction is the main processing stage

PDF text extraction requires significantly more processing time than the remaining parsing stages.

2. Text cleaning is lightweight

The text-cleaning stage completes in approximately 0.001 seconds.

3. Section detection is efficient

Section detection completes in approximately 0.001 seconds and successfully identifies the major resume sections.

4. Candidate extraction is efficient

Candidate extraction also requires approximately 0.001 seconds.

5. Repeated benchmarking provides a better baseline

A five-run benchmark produced an average PDF extraction time of approximately 0.48 seconds and an average complete parsing time of approximately 0.35 seconds. The repeated benchmark provides a more representative baseline than relying on a single execution.

Optimization Decision

No major optimization was applied during Day 18.

The current PDF extraction implementation is simple and functional. Optimization should only be introduced after testing alternative extraction approaches and confirming that extraction accuracy is not reduced.

Future optimization can focus on PDF extraction if larger documents or high-volume resume processing create a performance bottleneck.

Conclusion

Day 18 established a performance baseline for the Zecpath AI resume parsing pipeline.

The repeated benchmark demonstrated that the current parser can process the test resume in well under one second on average during repeated execution.

The parser correctly identifies the candidate information and major resume sections.

The main area for future performance optimization is PDF text extraction, while the remaining parsing stages are already lightweight and efficient.